

*Part A***Question 1:****a)**

By Definition 1.9 Chapter 13 p63 and Theorem 1.22 Chapter 13 p 64;

i.

An example of a strictly monotonic decreasing sequence is;

$$(x_n) = 1 + \frac{1}{n}, \text{ for } n \geq 1$$

It is decreasing because;

$$\begin{aligned} x_{n+1} - x_n &= \left(1 + \frac{1}{n+1}\right) - \left(1 + \frac{1}{n}\right) \\ &= \frac{1}{n+1} - \frac{1}{n} \\ &= \frac{n - (n+1)}{n(n+1)} \\ &= \frac{-1}{n(n+1)} < 0 \end{aligned}$$

for all $n \geq 1$

and,

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = 1 + 0 = 1$$

Since (x_n) is strictly monotonic decreasing and bounded below by 1, it converges, Theorem 1.22 HB Chapter 13 p64, and from the above we see that,

$$\lim_{n \rightarrow \infty} x_n = 1$$

ii.

An example of a strictly monotonic increasing sequence is;

$$(y_n) = -1 - \frac{1}{n}, \text{ for } n \geq 1$$

It is increasing because;

$$\begin{aligned}
 y_{n+1} - y_n &= \left(-1 - \frac{1}{n+1}\right) - \left(-1 - \frac{1}{n}\right) \\
 &= -\frac{1}{n+1} + \frac{1}{n} \\
 &= \frac{-n + (n+1)}{n(n+1)} \\
 &= \frac{1}{n(n+1)} > 0
 \end{aligned}$$

for all $n \geq 1$

and,

$$\lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} \left(-1 - \frac{1}{n}\right) = -1 - 0 = -1$$

Since (y_n) is strictly monotonic increasing and bounded above by -1, it converges, Theorem 1.22 HB Chapter 13 p64, and from the above we see that,

$$\lim_{n \rightarrow \infty} y_n = -1$$

b)

$$(z_n) = \begin{cases} x_n, & \text{if } n \text{ is odd} \\ y_n, & \text{if } n \text{ is even} \end{cases}$$

Using Definition 1.24 HB Chapter 13 p65, where (x_n) is the strictly monotonic decreasing sequence defined in part (a)(i) and (y_n) is the strictly monotonic increasing sequence defined in part (a)(ii). We can see that the subsequence of odd terms (z_{2n-1}) is equivalent to (x_n) and the subsequence of even terms (z_{2n}) is equivalent to (y_n) . Thus we have that;

$$\lim_{n \rightarrow \infty} z_{2n-1} = \lim_{n \rightarrow \infty} x_n = 1$$

and

$$\lim_{n \rightarrow \infty} z_{2n} = \lim_{n \rightarrow \infty} y_n = -1$$

Since the two subsequences converge to different limits, and by Proposition 1.25 HB Chapter 13 p65, the sequence (z_n) does not converge.

Question 2:

$$L = L_{a,\theta} = \{(x, a + x \tan \theta) : x \in \mathbb{R}\}$$

Where a is the y -intercept of the line and θ is the angle of inclination of the line to the positive x -axis.

$$X = L_{a,\theta} : a \in \mathbb{R} \text{ and } \theta \in (-\pi/2, \pi/2)$$

and the distance function;

$$\Delta(L_{a,\theta}, L_{b,\phi}) = |a - b| + |\tan \theta - \tan \phi|$$

for $L_{a,\theta}, L_{b,\phi} \in X$

a)

The distance between $L_{0,\pi/4}$ and $L_{1,-\pi/4}$;

$$\begin{aligned} \Delta(L_{0,\pi/4}, L_{1,-\pi/4}) &= |0 - 1| + |\tan \pi/4 - \tan -\pi/4| \\ &= 1 + |1 - (-1)| \\ &= 1 + |2| \\ &= 1 + 2 = 3 \end{aligned}$$

b)

By Definition 2.1, HB Chapter 14 p72,

i.

To show Δ satisfies (M1) non-negativity, we need to show that for all $(L_{a,\theta}, L_{b,\phi} \in X)$,

$$\Delta(L_{a,\theta}, L_{b,\phi}) \geq 0$$

$$\Delta(L_{a,\theta}, L_{b,\phi}) = |a - b| + |\tan \theta - \tan \phi|$$

As $\mathbf{a}, \mathbf{b} \in \mathbb{R}$ and $\theta \in (-\pi/2, \pi/2)$, both $|a - b| \geq 0$ and $|\tan \theta - \tan \phi| \geq 0$

Hence

$$\Delta(L_{\mathbf{a}, \theta}, L_{\mathbf{b}, \phi}) \geq 0$$

So, $\Delta(L_{\mathbf{a}, \theta}, L_{\mathbf{b}, \phi}) \geq 0$ and thus Δ is non-negative.

If

$$\begin{aligned} \Delta(L_{\mathbf{a}, \theta}, L_{\mathbf{b}, \phi}) &= |a - a| + |\tan \theta - \tan \theta| \\ &= 0 + 0 = 0 \end{aligned}$$

and

$$\Delta(L_{\mathbf{a}, \theta}, L_{\mathbf{b}, \phi}) = 0$$

then

$$|a - b| + |\tan \theta - \tan \phi| = 0$$

Since both terms are non-negative, they must both be zero, so

$$|a - b| = 0 \text{ and } |\tan \theta - \tan \phi| = 0$$

which implies

$$a = b \text{ and } \tan \theta = \tan \phi$$

and since $\theta, \phi \in (-\pi/2, \pi/2)$,

$$\theta = \phi$$

Thus, $\Delta(L_{\mathbf{a}, \theta}, L_{\mathbf{b}, \phi}) = 0$ if and only if $a = b$ and $\theta = \phi$, so (M1) is satisfied.

ii.

To show Δ satisfies (M2) symmetry, we need to show that for any $\Delta(\mathbf{a}, \mathbf{b}) = \Delta(\mathbf{b}, \mathbf{a})$

$$\Delta(L_{a,\theta}, L_{b,\phi}) = |a - b| + |\tan \theta - \tan \phi|$$

By the properties of absolute values, HB p8, $|a - b| = |b - a|$ and $|\tan \theta - \tan \phi| = |\tan \phi - \tan \theta|$

$$= |b - a| + |\tan \phi - \tan \theta|$$

$$= \Delta(L_{b,\phi}, L_{a,\theta})$$

Thus, $\Delta(L_{a,\theta}, L_{b,\phi}) = \Delta(L_{b,\phi}, L_{a,\theta})$ and (M2) is satisfied.

iii.

To show Δ satisfies (M3) triangle inequality, we need to show that for any $\Delta(a, c) \leq \Delta(a, b) + \Delta(b, c)$

$$\Delta(L_{a,\theta}, L_{c,\psi}) = |a - c| + |\tan \theta - \tan \psi|$$

By the triangle inequality for real numbers, HB p9,

$$\leq (|a - b| + |b - c|) + (|\tan \theta - \tan \phi| + |\tan \phi - \tan \psi|)$$

Since

$$|a - c| \leq |a - b| + |b - c|$$

And

$$|\tan \theta - \tan \psi| \leq |\tan \theta - \tan \phi| + |\tan \phi - \tan \psi|$$

$$= |a - b| + |\tan \theta - \tan \phi| + |b - c| + |\tan \phi - \tan \psi|$$

Thus, using the triangle inequality twice, $\Delta(L_{a,\theta}, L_{c,\psi}) \leq \Delta(L_{a,\theta}, L_{b,\phi}) + \Delta(L_{b,\phi}, L_{c,\psi})$ and (M3) is satisfied.

c)

$$\mathbf{B}_{\Delta}(\mathbf{L}_{0,0}, 1)$$

By Definition 2.6, HB Chapter 14 p72 for open balls in a metric space, we have that,

$$\begin{aligned}\mathbf{B}_{\Delta}(\mathbf{L}_{0,0}, 1) &= \{\mathbf{L}_{b,\theta} \in \mathbf{X} : \Delta(\mathbf{L}_{0,0}, \mathbf{L}_{b,\theta}) < 1\} \\ &= \{\mathbf{L}_{b,\theta} \in \mathbf{X} : |0 - b| + |\tan 0 - \tan \theta| < 1\} \\ &= \{\mathbf{L}_{b,\theta} \in \mathbf{X} : |b| + |0 - \tan \theta| < 1\} \\ &= \{\mathbf{L}_{b,\theta} \in \mathbf{X} : |b| + |\tan \theta| < 1\}\end{aligned}$$

d)

The sequence of lines;

$$(\mathbf{L}_n)_{n=1}^{\infty} \text{ given by } \mathbf{L}_n = \mathbf{L}_{-1/n, \pi/2 - 1/n}$$

is not Δ -convergent because, by Definition 3.2 HB Chapter 14 p73;

Using $\mathbf{L}_{b,\phi} \in \mathbf{X}$ Then

$$\Delta(\mathbf{L}_n, \mathbf{L}_{b,\phi}) = \left| -\frac{1}{n} - b \right| + \left| \tan\left(\frac{\pi}{2} - \frac{1}{n}\right) - \tan \phi \right|$$

As $n \rightarrow \infty$, we have $-1/n \rightarrow 0$, so

$$\left| -\frac{1}{n} - b \right| \rightarrow |b|$$

Also, as $n \rightarrow \infty$, we have

$$\tan\left(\frac{\pi}{2} - \frac{1}{n}\right) \rightarrow +\infty$$

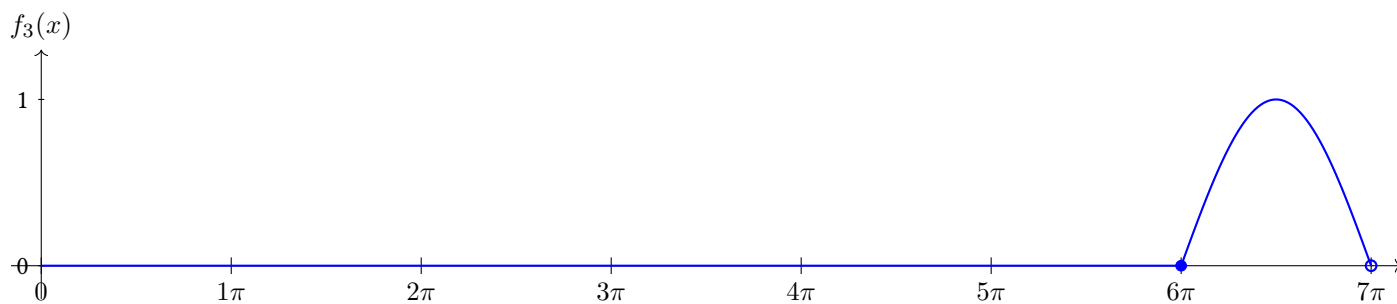
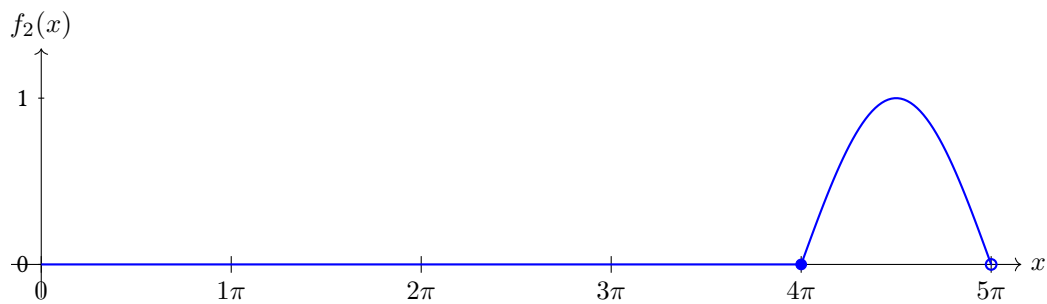
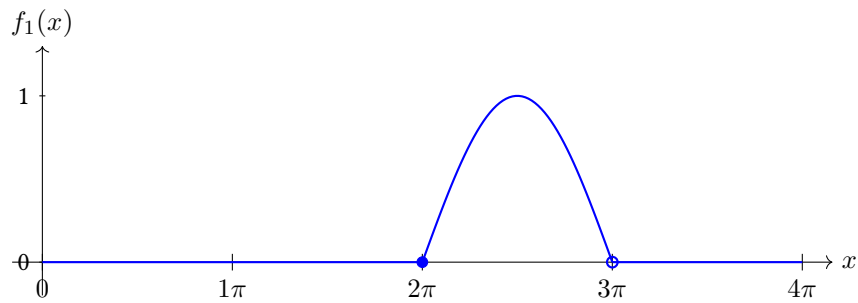
and since $\tan \phi$ is finite for all $\phi \in (-\pi/2, \pi/2)$,

$$\left| \tan\left(\frac{\pi}{2} - \frac{1}{n}\right) - \tan \phi \right| \rightarrow \infty.$$

Hence $\Delta(\mathbf{L}_n, \mathbf{L}_{b,\phi}) \rightarrow \infty$, and so $(\mathbf{L}_n)$ is not Δ -convergent.

Question 3:

$$f_n(x) = \begin{cases} \sin(\pi x), & \text{if } 2n \leq x < (2n+1), \\ 0, & \text{otherwise.} \end{cases}$$

a)**b)**

To test for pointwise limits we can take $x = 2.5$ as a constant value.

When $n = 1$ the interval is $[2, 3)$

$$f_1(2.5) = \sin(2.5\pi) = 1$$

When $n = 2$ the interval is $[4, 5)$

$$f_2(2.5) = 0$$

When $n = 3$ the interval is $[6, 7)$

$$f_3(2.5) = 0$$

$\vdots$

When $n = k$ the interval is $[2k, 2k + 1)$

$$f_k(2.5) = 0$$

By the Archimedean property, for any value of x there exists a natural number $N \in \mathbb{N}$ such that $2N > x$.

Then for any $n > N$, $2n > x$ and so $f_n = 0$.

It follows that $f_n(x) \rightarrow 0$.

And so we conclude that the sequence (f_n) has a pointwise limit function f , given by;

$$f(x) = 0 \text{ for all } x \in \mathbb{R}$$

c)

Using Definition 4.4 Chapter 15 p 78 we can test for uniform convergence.

$$\begin{aligned} \forall n \in \mathbb{R}, |f_n(x) - f(x)| &= |f_n(x) - 0| \\ &= |f_n(x)| \\ &= \begin{cases} |\sin(\pi x)|, & \text{if } 2n \leq x < (2n + 1), \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Hence

$$\begin{aligned}\sup_{x \in \mathbb{R}} |f_n(x) - f(x)| &= \sup_{x \in \mathbb{R}} |f_n(x)| \\ &= \sup_{x \in [2n, 2n+1)} |\sin(\pi x)| \\ &= 1\end{aligned}$$

It follows that f_n is not a null sequence of functions, and so f_n does not converge uniformly to f as $n \rightarrow \infty$

Question 4:

$$F : C[0, 1] \rightarrow C[0, 1]$$

$$F(f)(t) = \int_t^1 (1 - x^4) f(x) \, dx \text{ for each } f \in C[0, 1] \text{ and } t \in [0, 1]$$

Let $f \in C[0, 1]$ and $s, t \in [0, 1]$ with $s \leq t$.

Let $f, g \in C[0, 1]$ and $t \in [0, 1]$.

$$\begin{aligned} |F(f)(t) - F(g)(t)| &= \left| \int_t^1 (1 - x^4) f(x) \, dx - \int_t^1 (1 - x^4) g(x) \, dx \right| \\ &= \left| \int_t^1 (1 - x^4) (f(x) - g(x)) \, dx \right| \\ &\leq \int_t^1 |(1 - x^4)(f(x) - g(x))| \, dx \\ &\leq \int_t^1 (1 - x^4) |f(x) - g(x)| \, dx \end{aligned}$$

Since $0 \leq (1 - x^4) \leq 1 \forall x \in [0, 1]$

$$\leq \int_t^1 |f(x) - g(x)| \, dx$$

By Definition 5.1 HB Chapter 15 p79

$$\leq d_{\max}(f, g) \forall x \in [0, 1]$$

Hence

$$\begin{aligned} &\leq \int_t^1 d_{\max}(f, g) \, dx = d_{\max}(f, g)(1 - t) \\ &\leq d_{\max}(f, g) \end{aligned}$$

Taking the supremum over $t \in [0, 1]$ on the left hand side

$$= \sup_{t \in [0, 1]} |F(f)(t) - F(g)(t)| \leq d_{\max}$$

Hence F is a d_{max}, d_{max} -Lipschitz function on $C[0, 1]$. Thus F is continuous by Theorem 3.2 Chapter 15 p76.

Question 5:

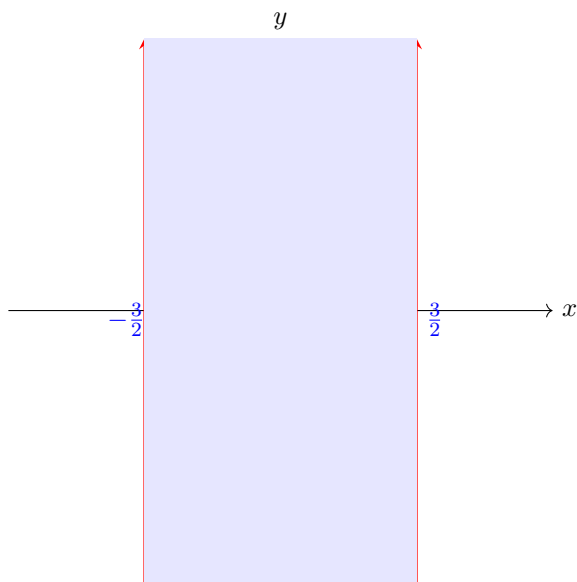
$$A = \{(x, y) \in \mathbb{R}^2 : 2|x| \leq 3\}$$

$$B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}$$

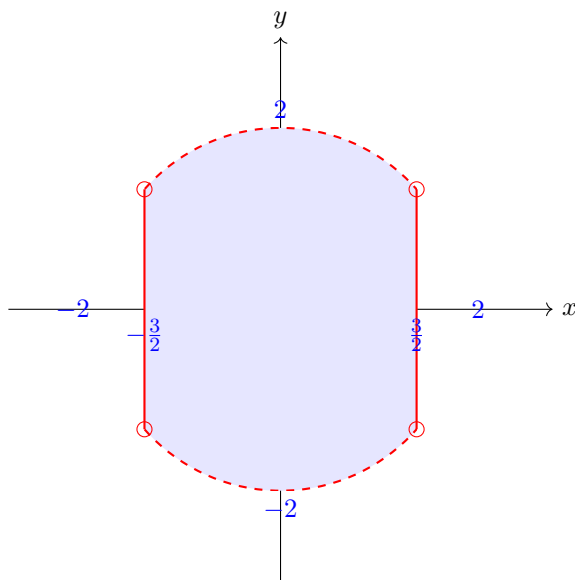
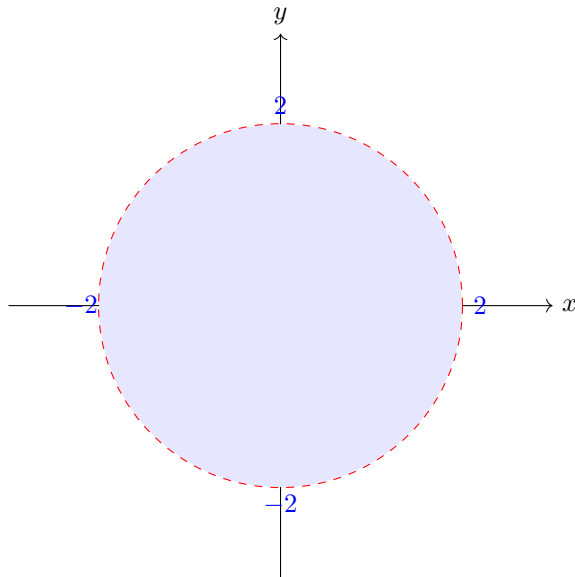
$$C = A \cap B$$

a)

Sketch of set (A):



Sketch of set (B):



b)

i.

By Definition 3.6 HB Chapter 16 p84 ,

let $x = \frac{3}{2}$ and solve for y in set B's inequality:

$$x^2 + y^2 < 4$$

;

then $y = \sqrt{2^2 - (\frac{3}{2})^2} = \sqrt{\frac{7}{4}} = \frac{\sqrt{7}}{2}$. Thus, one closure point is $(\frac{3}{2}, \frac{\sqrt{7}}{2})$.

ii.

Closure of set C is:

$$C1 = \{(x, y) \in \mathbb{R}^2 : 2|x| \leq 3 \text{ and } x^2 + y^2 \leq 4\}$$

Interior of set C is:

$$Int = \{(x, y) \in \mathbb{R}^2 : 2|x| < 3 \text{ and } x^2 + y^2 < 4\}$$

Boundary of set C is:

$$Bd = \{(x, y) \in \mathbb{R}^2 : (2|x| = 3 \text{ and } x^2 + y^2 \leq 4) \text{ or } (2|x| \leq 3 \text{ and } x^2 + y^2 = 4)\}$$

c)

Given the 'mixed' metric d ;

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + d_0(y_1, y_2)$$

With d_0 the discrete metric on $\mathbb{R}$.

By Definition 2.3 HB Chapter 14 p72

$$d_0(y_1, y_2) = \begin{cases} 0, & \text{if } y_1 = y_2, \\ 1, & \text{if } y_1 \neq y_2. \end{cases}$$

Consider

$$C = \{(x, y) \in \mathbb{R}^2 : 2|x| \leq 3 \text{ and } x^2 + y^2 < 4\}$$

Take (x_0, y_0) to satisfy

$$2|x_0| \leq 3 \text{ and } x_0^2 + y_0^2 \leq 4$$

Then if $x_0^2 + y_0^2 < 4$ then $x_0^2 + y_0^2 \in C$ and therefore a closure point.

And if $x_0^2 + y_0^2 = 4$ and $2|x_0| \leq 3$ then it is possible to choose a sequence (x_n, y_0) with $x_n^2 + y_0^2 < 4$ and $x_n \rightarrow x_0$. As y stays constant,

$$d((x_n, y_0), (x_0, y_0)) = |x_n - x_0| \rightarrow 0$$

Hence boundary points are also closure points and the closure of C is:

$$C1 = \{(x, y) \in \mathbb{R}^2 : 2|x| \leq 3 \text{ and } x^2 + y^2 \leq 4\}$$

Question 6:

$$D = \left\{ x \in \mathbb{R} : x = \frac{m}{p} \text{ for some } m \in \mathbb{N} \text{ and } p \text{ prime} \right\}$$

a)

$m \in \mathbb{N}$ is countable, Definition 6.4 HB Chapter 16 p87, and $p \in \mathbb{N}$ is countable. Hence $D \in \mathbb{Q}$, is countable, Theorem 6.6 HB Chapter 16 p87.

b)**i.***Proof.*

For

$$y \geq 0$$

and

$$r > 0$$

Hence

$$y + r > 0$$

Let p be a prime number, in the interval

$$(yp, p(y+r))$$

The length of this interval is

$$p(y+r) - yp = pr$$

As $r > 0$, and primes are unbounded, there exists a prime p such that

$$pr > 1$$

Hence

$$p(y+r) - yp = pr > 1$$

Therefore contains at least one integer $m \in \mathbb{N}$

Thus

$$yp < m < p(y+r)$$

Dividing by p gives

$$y < \frac{m}{p} < y + r$$

$$\text{With } \frac{m}{p} \in D$$

Therefore, there exists

$$x = \frac{m}{p} \in D$$

such that

$$y < x < y + r$$

□

ii.

Proof.

To show the D is dense in $[0, \infty)$, we need to show that for any

$$a, b \in [0, \infty)$$

such that

$$a < b$$

there exists

$$x \in D$$

such that

$$a < x < b$$

Let

$$r = b - a > 0$$

From part (i), we know that there exists

$$x \in D$$

such that

$$a < x < a + r$$

But

$$a + r = a + (b - a) = b$$

Thus

$$a < x < b$$

and so D is dense in $[0, \infty)$.

□

Question 7:

I have watched the tutorial but forgot the date. I have emailed my tutor and was advised to write this as a proof of enagement.

*Part B***Question 8:**

Question 8 from section B

Let R be the ring $\mathbb{Z}[\sqrt{-13}]$

a)

$$A = \{s + t\sqrt{-13} : s, t \in \mathbb{Z}, s - t \equiv 0 \pmod{7}\}$$

As $s \equiv t \pmod{7}$ there exists an integer k such that,

$$s = t + 7k$$

Hence,

$$\begin{aligned} s + t\sqrt{-13} &= (t + 7k) + t\sqrt{-13} \\ &= t(1 + \sqrt{-13}) + 7k \end{aligned}$$

So

$$A\langle 7, 1 + \sqrt{-13} \rangle$$

By Definition 2.1 HB Chapter 17 p94,

To Show A is ideal, it must be closed under multiplication by any $r \in R$.
So we need the product of the generator and $\sqrt{-13}$ to stay in A .

$$\sqrt{-13}(1 + \sqrt{-13}) = \sqrt{-13} - 13$$

since

$$-13 \equiv 1 \pmod{7}$$

$$-13 = 1 - 14$$

$$\begin{aligned} \sqrt{-13} - 13 &= (\sqrt{-13} + 1) - 14 \\ &= (\sqrt{-13} + 1) + 7(-2) \end{aligned}$$

Since $(\sqrt{-13} + 1) \in A$ and $14 = 2(7) \in A$ it follows that A is closed under multiplication by any element of R and hence is an ideal.

b)

$$B = \{u + v\sqrt{-13} : u, v \in \mathbb{Z}, u + v \equiv 0 \pmod{7}\}$$

As $u = -v \pmod{7}$ there exists an integer k such that,

$$u = -v + 7k$$

Hence,

$$\begin{aligned} u + v\sqrt{-13} &= (-v + 7k) + v\sqrt{-13} \\ &= v(-1 + \sqrt{-13}) + 7k \end{aligned}$$

So

$$B\langle 7, -1 + \sqrt{-13} \rangle$$

because -1 is a unit in $\mathbb{Z}[\sqrt{-13}]$, therefore we can multiply the second generator by -1 and it will still generate the same ideal.

$$B\langle 7, 1 - \sqrt{-13} \rangle$$

Hence,

$$A\langle 7, 1 + \sqrt{-13} \rangle$$

and

$$B\langle 7, 1 - \sqrt{-13} \rangle$$

By Lemma 3.9 HB Chapter 17 p97, the product of two ideals A and B

$$AB\langle 7 \cdot 7, 7(1 - \sqrt{-13}), 7(1 + \sqrt{-13}), (1 + \sqrt{-13})(1 - \sqrt{-13}) \rangle$$

$$\langle 49, 7(1 - \sqrt{-13}), 7(1 + \sqrt{-13}), 14 \rangle$$

Since the $\text{hcf}(14, 49) = 7$ and $7(1 - \sqrt{-13})$ and $7(1 + \sqrt{-13})$ are multiples of 7 and,

$$7 = 49 - 3(14) \in AB$$

$$\langle 7 \rangle \subseteq AB$$

hence

$$AB\langle 7 \rangle$$

Question 9 from section B **Question 9:**

$$B_d((0, 0), 1), B_d[(0, 0), 1], \text{ and } S_s((0, 0), 1)$$

By Defintion 2.6 HB Chapter 14 p 72

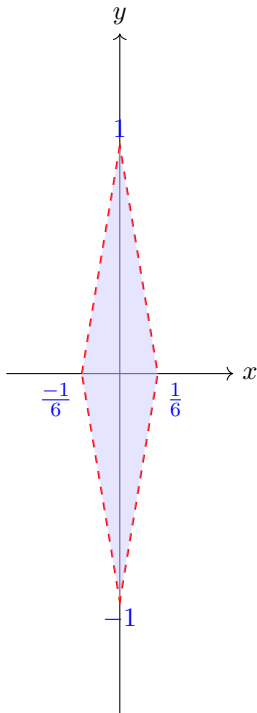
a)

The open ball on metric

$$d((a_1, a_2), (b_1, b_2)) = 6|b_1 - a_1| + |b_2 - a_2|$$

is given by

$$\begin{aligned} B_d((0, 0), 1) &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : 6|x - 0| + |y - 0| < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : 6|x| + |y| < 1\} \end{aligned}$$

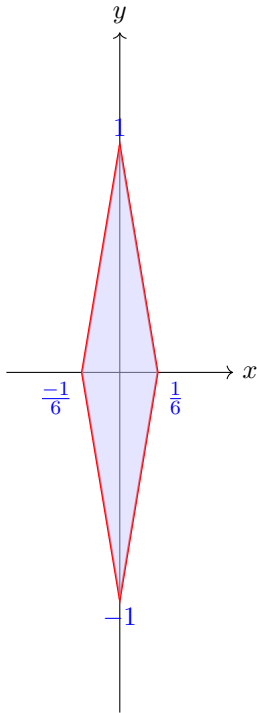


The closed ball on metric

$$d((a_1, a_2), (b_1, b_2)) = 6|b_1 - a_1| + |b_2 - a_2|$$

is given by

$$\begin{aligned}
 B_d[(0,0),1] &= \{(x,y) \in \mathbb{R}^2 : d((x,y), (0,0)) \leq 1\} \\
 &= \{(x,y) \in \mathbb{R}^2 : 6|x-0| + |y-0| \leq 1\} \\
 &= \{(x,y) \in \mathbb{R}^2 : 6|x| + |y| \leq 1\}
 \end{aligned}$$

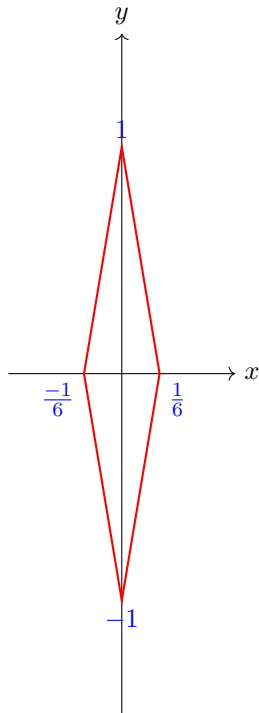


The sphere on metric

$$d((a_1, a_2), (b_1, b_2)) = 6|b_1 - a_1| + |b_2 - a_2|$$

is given by

$$\begin{aligned}
 S_s((0,0),1) &= \{(x,y) \in \mathbb{R}^2 : d((x,y), (0,0)) = 1\} \\
 &= \{(x,y) \in \mathbb{R}^2 : 6|x-0| + |y-0| = 1\} \\
 &= \{(x,y) \in \mathbb{R}^2 : 6|x| + |y| = 1\}
 \end{aligned}$$



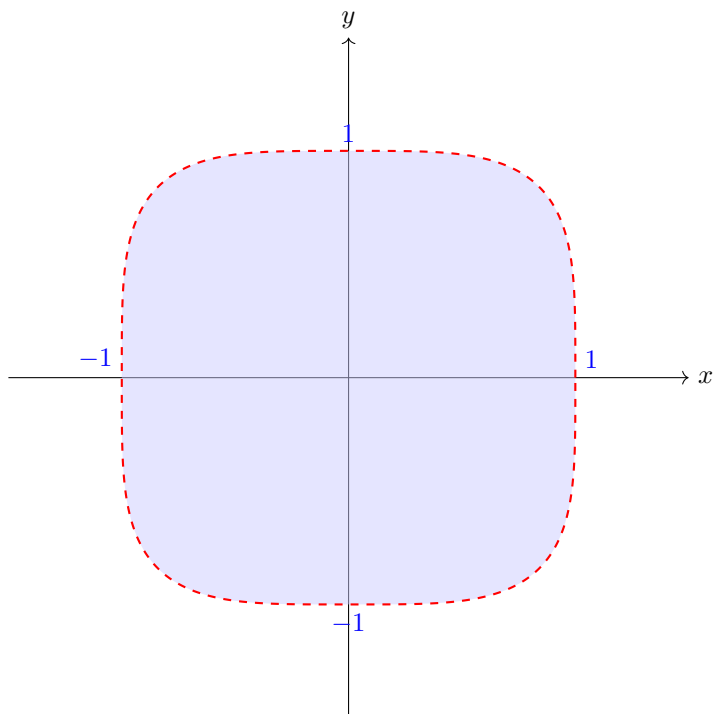
b)

The open ball on metric

$$d((a_1, a_2), (b_1, b_2)) = (|b_1 - a_1|^4 + |b_2 - a_2|^4)^{\frac{1}{4}}$$

is given by

$$\begin{aligned} B_d((0, 0), 1) &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x - 0|^4 + |y - 0|^4)^{\frac{1}{4}} < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x|^4 + |y|^4)^{\frac{1}{4}} < 1\} \end{aligned}$$

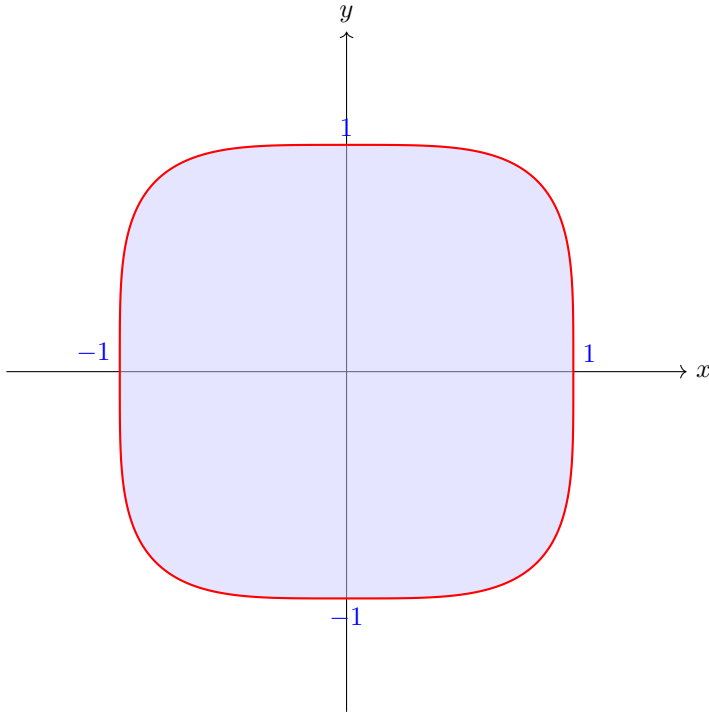


The closed ball on metric

$$d((a_1, a_2), (b_1, b_2)) = (|b_1 - a_1|^4 + |b_2 - a_2|^4)^{\frac{1}{4}}$$

is given by

$$\begin{aligned} B_d[(0, 0), 1] &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) \leq 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x - 0|^4 + |y - 0|^4)^{\frac{1}{4}} \leq 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x|^4 + |y|^4)^{\frac{1}{4}} \leq 1\} \end{aligned}$$

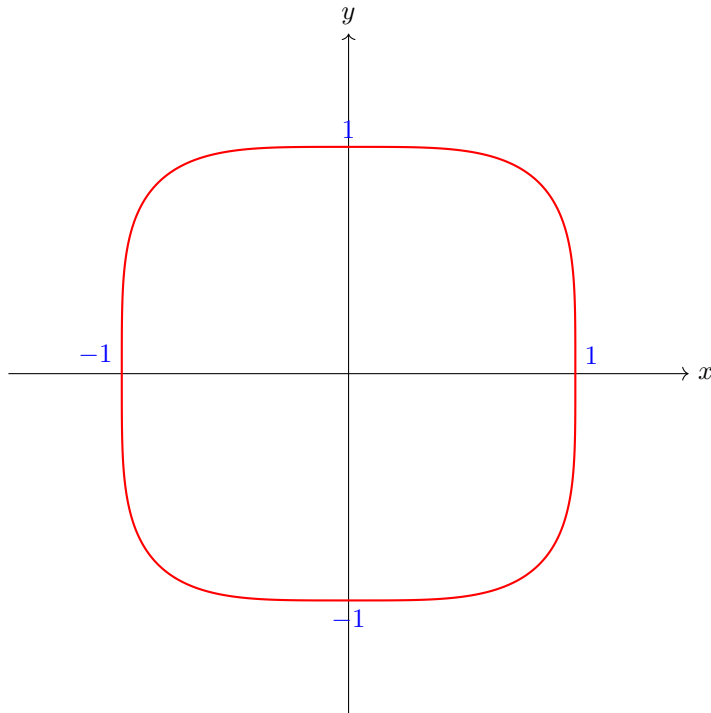


The sphere on metric

$$d((a_1, a_2), (b_1, b_2)) = (|b_1 - a_1|^4 + |b_2 - a_2|^4)^{\frac{1}{4}}$$

is given by

$$\begin{aligned} S_s((0, 0), 1) &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) = 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x - 0|^4 + |y - 0|^4)^{\frac{1}{4}} = 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x|^4 + |y|^4)^{\frac{1}{4}} = 1\} \end{aligned}$$



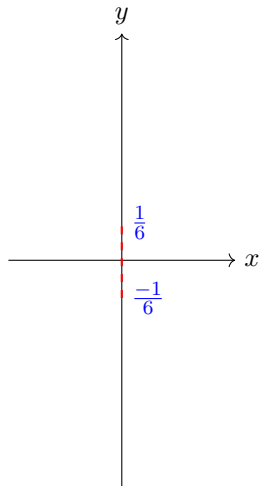
c)

The open ball on metric

$$d((a_1, a_2), (b_1, b_2)) = d_0(a_1, b_1) + 6|b_2 - a_2|$$

is given by

$$\begin{aligned} S_s((0, 0), 1) &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : d_0(x, 0) + 6|y - 0| < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : d_0(x, 0) + 6|y| < 1\} \end{aligned}$$

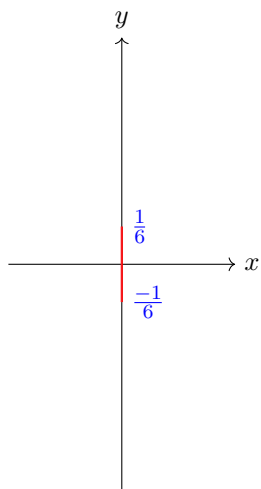


The closed ball on metric

$$d((a_1, a_2), (b_1, b_2)) = d_0(a_1, b_1) + 6|b_2 - a_2|$$

is given by

$$\begin{aligned} B_d[(0, 0), 1] &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) \leq 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : d_0(x, 0) + 6|y - 0| \leq 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : d_0(x, 0) + 6|y| \leq 1\} \end{aligned}$$



The sphere on metric

$$d((a_1, a_2), (b_1, b_2)) = d_0(a_1, b_1) + 6|b_2 - a_2|$$

is given by

$$\begin{aligned} S_s((0,0),1) &= \{(x,y) \in \mathbb{R}^2 : d((x,y), (0,0)) = 1\} \\ &= \{(x,y) \in \mathbb{R}^2 : d_0(x,0) + 6|y-0| = 1\} \\ &= \{(x,y) \in \mathbb{R}^2 : d_0(x,0) + 6|y| = 1\} \end{aligned}$$

